Practical 8

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int time[3];
    int i,j,k;

    int total_pages = 12;
    int frames_count = 3;
    int hit = 0, fault = 0;
    int counter = 0;

    for(i=0;i<frames_count;i++){
        frames[i] = -1;
    }

    for(i=0;i<total_pages;i++){
        int found = 0;

        for(j=0;j<frames_count;j++){
            if(frames[j] == pages[i]){
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found){
            int pos = 0;

            if(fault < frames_count){
                pos = fault;
            }
            else{
                int min = time[0];
                pos = 0;
                for(k=1;k<frames_count;k++){
                    if(time[k] < min){
                        min = time[k];
                        pos = k;
                    }
                }
            }

            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
            fault++;
        }
    }
```

```
            fault++;
        }
    }

    printf("Total Page References = %d\n", total_pages);
    printf("Page Hits = %d\n", hit);
    printf("Page Faults = %d\n", fault);

    float hit_percent = (hit * 100.0) / total_pages;
    float fault_percent = (fault * 100.0) / total_pages;

    printf("Hit Percentage = %.2f%%\n", hit_percent);
    printf("Fault Percentage = %.2f%%\n", fault_percent);

    return 0;
}
```

Output:

```
M ~

ladde@Hashirama MSYS ~
$ nano practical8.c

ladde@Hashirama MSYS ~
$ gcc practical8.c -o practical8

ladde@Hashirama MSYS ~
$ ./practical8
Total Page References = 12
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

ladde@Hashirama MSYS ~
$
```